

1. Imagine a simplified ocean (we are thinking about a surface ocean isolated from the deep ocean) where surface seawater has $S=35$ and $T = 25^{\circ}\text{C}$. Assume that the alkalinity of this system is determined entirely by the carbonate system, i.e. $A_T=A_c$, where,

$$A_c = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

In class we discussed how this definition leaves out $[\text{H}^+]$, among other species, and therefore is an imperfect definition of Alkalinity. However, for the global mean surface seawater, $[\text{HCO}_3^-]$ makes up 76.8% of Alkalinity and $[\text{CO}_3^{2-}]$ makes up 18.8% (Sarmiento and Gruber, 2006), while $[\text{H}^+]$ only makes up $\sim 0.2\%$ of Alkalinity, so using carbonate alkalinity (A_c) is a reasonable approximation that lets us solve carbonate problems by hand rather than using a program.

Let us also assume that $\text{DIC} = [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$, i.e. there is very little $[\text{CO}_2]^*$ relative to the other carbonate species.

$$A_c = 2140 \mu\text{eq kg}^{-1}$$

$$\text{DIC} = 1900 \mu\text{mol kg}^{-1}$$

$$K_{\text{H},\text{CO}_2} = 0.0284 \text{ mol kg}^{-1} \text{ atm}^{-1} \text{ at } 25^{\circ}\text{C}$$

$$K_1' = 10^{-5.85}$$

$$K_2' = 10^{-8.97}$$

- a. Assuming this ocean is isolated from the atmosphere (say it is entirely covered by sea ice, despite the 25°C SST), determine the pH, the distribution of the carbonate species $[\text{HCO}_3^-]$ and $[\text{CO}_3^{2-}]$ in $\mu\text{mol kg}^{-1}$, and the fCO_2 in ppm or μatm .
- b. Now consider what happens when this simplified surface ocean is exposed to the present-day atmosphere ($\text{fCO}_2 = 425 \text{ ppm}$ – *I hate that I have to regularly update this number*). The carbonate system will reach a new equilibrium with atmospheric fCO_2 . Calculate the new pH and the new concentrations of the carbonate species $[\text{HCO}_3^-]$ and $[\text{CO}_3^{2-}]$. Despite our abbreviated definition of alkalinity, keep in mind that alkalinity represents the charge balance of the ocean, so even though adding or removing CO_2 will change the relative amounts of bicarbonate and carbonate ion, it won't change alkalinity by an appreciable amount.
- c. Now consider the effect of primary production on the simplified ocean in part (a) i.e. capped once again by sea ice. Imagine two scenarios where primary production draws down DIC by $200 \mu\text{mol kg}^{-1}$. In the first case, diatoms account for all primary production (no CaCO_3 produced). In the second case, coccolithophores account for all primary production (assume a CaCO_3 to Organic C ratio of 1). Assume in your consideration of alkalinity that 50% of primary production involved NO_3^- uptake. What is the fCO_2 of the surface ocean in each case?
- d. Explain the implications of your results from part (c) in the context of ocean acidification. Consider possible changes to CaCO_3 producers and existing CaCO_3 (i.e. in reefs / sediments) and possible feedbacks to the carbon cycle. Would fostering

CaCO_3 producers be a viable source of geoengineering to slow the increase in atmospheric CO_2 ?

2. The diagram below is created from nitrate measurements made in sediment at the ocean floor. The water temperature at this depth is 2°C .
 - a) Draw the different redox zones present in this sediment based on the nitrate concentration profile below.
 - b) Calculate the diffusive fluxes at the sediment-water interface and at 2cm depth, assuming a porosity of 0.8 and a diffusion coefficient of nitrate in seawater (at 2°C) of at $330\text{ cm}^2/\text{y}$. Indicate the direction of the flux.
 - c) Estimate the amount of organic matter respired per year needed to yield this nitrate profile. Assume Redfield stoichiometry is maintained during respiration.



